

				Title	Correction and addition of the description for FANUC Robot series R-30iB/R-30iB Mate/R-30iB Plus/R-30iB Mate Plus CONTROLLER OPERATOR'S MANUAL (Collaborative Robot Function)					
					Draw No.	B-83744EN/02-03				
02	2019.4.8	Wang	Correction in P6 and P8			FANUC CORPORATION			Sheet	1／12
ED	Date	Design	Description							
	Date	2019.1.31	Design	Naitou, Wang	Approve	Arita				

Change the following items in section 2.2. (Change is underlined)

Limit 1 to 4 (configurable)

Set the external force limit 1 to 4 between 0[N] and 150[N], inclusive. 0[N] means disable and you cannot set the limit 1 to 0[N].

When the contact stop function is enabled, “@” is displayed as following.

Limit 1:	150.00[N]	---[0]	OK
@Limit 2:	90.00[N]	---[0]	OK
Limit 3:	0.00[N]	---[0]	OK
Limit 4:	0.00[N]	---[0]	OK

This means the active force limit. The active force limit is minimum limit in available limits. When the external force exceeds the active force limit, the robot will stop. To disable the force limit, set the limit to 0[N] or the corresponding disabling input to ON. You can disable the force limit to change the active force limit dynamically in a program. In the example of this screen, the limit 1 and 2 are enabled, then @ is attached to the minimum limit 2. The limit 3 and 4 is set 0[N], then they are disabled.

When you push [NEXT] key and F5[DETAIL] on each limit item, detail setting screen is displayed.

Limit 1:	150.00[N]	OK
Disabling input	---[0]	OK
Payload Error Margin	60.00[N]	OK

Payload Error Margin (configurable)

Set Payload Error Margin. When steady load applied to the robot exceeds Payload Error Margin, SYST-325 “Payload error is detected” will be posted, and the robot will stop.

DCS					
Collaborative robot					
Speed Limit					
Collaborative Speed:		250 [mm/s]	OK		
Speed Clamping:		ENABLE			
Disabling Input:		DI[0]			
Max Speed:		250 [mm/s]			
[TYPE]	CONFIRM	PEAKCLR		UNDO	

Speed limit

Collaborative Speed (configurable)

This item is displayed in the software version 7DC3/28 or later. When the contact stop is enable, the TCP speed or the elbow (joint position between J2 arm and J3 arm) speed of the collaborative robot exceeds the speed limit, “SYST-323 Collaborative speed limit(TCP)” or “SYST-359 Collaborative speed limit(elbow)” will be posted, and the robot will stop.

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				Draw No.	B-83744EN/02-03					
02	2019.4.8	Wang	Correction in P6 and P8			FANUC CORPORATION			Sheet	2／12
ED	Date	Design	Description							
	Date	2019.1.31	Design	Naitou, Wang	Approve					

**WARNING**

If the speed limit is set to inappropriate value, the force at the time of contact increases, and a serious personal injury could result. Risk assessment for the whole robot system is required to determine appropriate speed limit. Refer to followings about the risk according to the speed.

CR-35*i*A: MECHANICAL UNIT OPERATOR'S MANUAL(B-83734EN), Section 3.6 THE CHARACTERISTIC OF COLLABORATIVE ROBOT AND LIMITATIONS AND USAGE NOTES

CR-4*i*A, CR-7*i*A, CR-7*i*A/L: MECHANICAL UNIT OPERATOR'S MANUAL(B-83774EN), Section 3.6 THE CHARACTERISTIC OF COLLABORATIVE ROBOT AND LIMITATIONS AND USAGE NOTES

CR-15*i*A: MECHANICAL UNIT OPERATOR'S MANUAL(B-84054EN), Section 3.6 THE CHARACTERISTIC OF COLLABORATIVE ROBOT AND LIMITATIONS AND USAGE NOTES

Speed Clamping, Disabling input (configurable)

ENABLE or DISABLE for speed clamping of collaborative speed is displayed. It is ENABLE by default. When you set Disabling input and turn signal to ON, it will turn to DISABLE.

Max Speed (configurable)

When you want to teach higher speed more than default, set this item to the maximum speed. Refer to Section 7.2 about procedure to switch the speed according to the conditions.

DCS					
Collaborative robot					
Use Payload Comp:		DISABLE		OK	
Enabling input:		ON [0]		OK	
Alarm in disabled:		DISABLE			
[TYPE]	CONFIRM	PEAKCLR		UNDO	

Use Payload Comp (configurable)

This item is displayed in the software version 7DC3/33 or later. Use when same payload acts on the robot every time during robot move. Refer to Section 7.5.

Enabling input (configurable)

When the set signal is ON, the compensation value is used, and when it is OFF, the compensation value is not used.

Alarm in disabled (configurable)

When this item is enabled, you use the KAREL programs for compensation with Use Payload Comp" disabled or Enabling input OFF, the robot will stop with alarm.

								Title	Correction and addition of the description for FANUC Robot series R-30 <i>i</i> B/R-30 <i>i</i> B Mate/R-30 <i>i</i> B Plus/R-30 <i>i</i> B Mate Plus CONTROLLER OPERATOR'S MANUAL (Collaborative Robot Function)			
									Draw No.	B-83744EN/02-03		
02	2019.4.8	Wang	Correction in P6 and P8				FANUC CORPORATION			Sheet	3／12	
ED	Date	Design	Description									
	Date	2019.1.31	Design	Naitou, Wang	Approve	Arita						

PyldComp during Escp (configurable)

7.5.3 Push to Escape with Variable Payload Compensation

In the software version 7DC3/47, 7DF1/17 or later, Push to Escape can be enabled when variable payload compensation is enabled.

Corrected in version 02



CAUTION

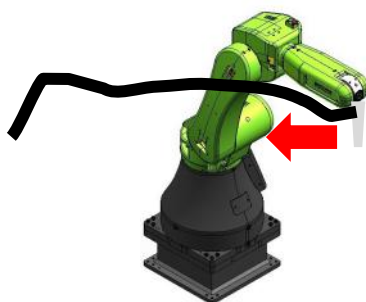
When the variable payload compensation is enabled and Push to Escape is enabled, the mismatch between recorded payload and actual payload causes Push to Escape movement with nobody pushing.

CONDITION

When [Check during moving] in [Auto status check] is enabled, [Push to Escape] cannot be enabled.

SETTING

- In Collaborative Robot screen, when both [Use Payload Comp] and [Push to Escape] are enabled, the message “Mismatch between the payload recorded by payload comp and the actual payload might cause Push to Escape movement with nobody pushing. Do you enable Push to Escape?” is displayed.
 - When [No] is selected, [Push to Escape] becomes disabled.
- When [Yes] is selected, [Push to Escape] becomes enabled, and the message “In case the payload remains during the escape, PyldComp during escp should be enabled. Do you enable PyldComp during escp?” is displayed.
 - In the case the payload still keeps even after robot position is changed by Push to Escape, select [Yes]. For example, when a cable is attached to the robot, payload by tension of cable keeps even if the robot position is changed. When [Yes] is selected, [PyldComp during Escp] becomes enabled and Compensation value keeps when Push to Escape is performed in PLAY mode.



CAUTION

When [PyldComp during Escp] is enabled and Robot position is changed by Push to Escape and the payload on the robot is changed, Push to Escape might be performed with nobody pushing because of mismatch between the recorded payload and the actual payload.

- In the case the payload will not work after Push to Escape, select [No]. For example, when the robot works while touching workpiece, the robot leaves workpiece by Push to escape, payload will not apply. When [No] is selected, [PyldComp during Escp] becomes disabled and the compensation value becomes 0 when Push to Escape is performed in PLAY mode. After restarting program, the compensation value will be back when the robot returns on original path.

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**CAUTION**

When [PyldComp during Escp] is disabled, if the payload keeps after Push to Escape, the payload might cause Push to Escape.

- [PyldComp during Escp] is also configurable in Collaborative Robot screen.

				Title	Correction and addition of the description for FANUC Robot series R-30iB/R-30iB Mate/R-30iB Plus/R-30iB Mate Plus CONTROLLER OPERATOR'S MANUAL (Collaborative Robot Function)					
				Draw No.	B-83744EN/02-03					
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ED	Date	Design	Description							
	Date	2019.1.31	Design	Naitou, Wang	Approve					

Section 7.7 is added as following

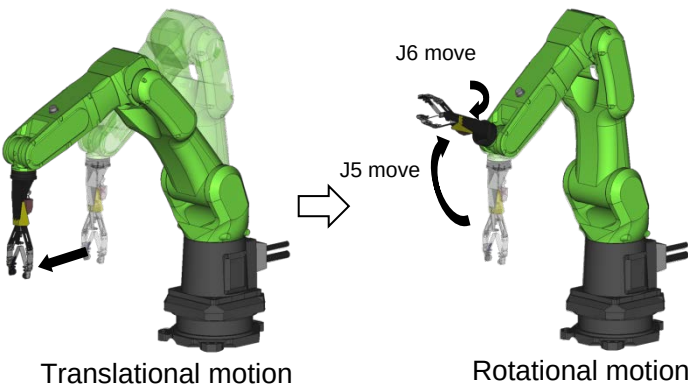
7.7 PAYLOAD IDENTIFICATION

Corrected in version 02

This function is available on the software version 7DC3/47, 7DF1/17 or after

This function identifies the mass of the payload and the center of gravity automatically by movement of collaborative robot.

Payload inertia is not identified with this function.



Execute payload identification

- 1 Press [MENU] key.
- 2 Select [SYSTEM].
- 3 Press F1, [TYPE].
- 4 Select [Motion].

MOTION PERFORMANCE				
Group1			1/10	
No.	PAYLOAD[kg]		Comment	
1	0.00	[	]	
2	0.00	[	]	
3	0.00	[	]	
4	0.00	[	]	
5	0.00	[	]	
6	0.00	[	]	
7	0.00	[	]	
8	0.00	[	]	
9	0.00	[	]	
10	0.00	[	]	
Active PAYLOAD number = 0				
[TYPE]	GROUP	DETAIL	ARMLoad	SETIND >

- 5 Press [NEXT], then press F2[IDENT]. The payload identification screen will be displayed.

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									Draw No.	B-83744EN/02-03			
02	2019.4.8	Wang	Correction in P6 and P8					FANUC CORPORATION			Sheet	9／12	
ED	Date	Design	Description										
	Date	2019.1.31	Design	Naitou, Wang	Approve	Arita							

- 1 When the payload identification is complete, “*****” of “1 PAYLOAD ESTIMATION *****” changes to [DONE]. However if it is aborted for some reason, such as Emergency Stop, it changes to [ABORT]. The payload are not estimated, so do not [APPLY] and execute the estimation again.
- 2 If the payload identification fails, any or all of the applied results (payload, center of gravity) would be 0.

- ### 7.7.1 Add positions of payload identification

Add positions of payload identification

- screen will be displayed.

MOTION/PAYLOAD ID		
		1/6
Base position	<RECORDED>	
Add position 1	< UNINIT >	<DISABLED>
Add position 2	< UNINIT >	<DISABLED>
Add position 3	< UNINIT >	<DISABLED>
Add position 4	< UNINIT >	<DISABLED>
Add position 5	< UNINIT >	<DISABLED>

Press F5 to confirm the position

[TYPE]	MOVE_TO	RECORD		POS	
----------	---------	--------	--	-----	--

- 2 Set the add position.
Add position should be set the same wrist posture, the same UT No. and the same UF No. with base position.
Move the robot to the desired position by jogging, then hold down the [SHIFT] key and press F3[RECORD] to record the position.
The status will change from [DISABLED] to [ENABLED] when add position recorded. Please change it to [DISABLED] if you do not use it.
By pressing F5[POS] shows the position information of add position recorded. The position information (XYZ) can be modified by entering their values directly.
Hold down the [SHIFT] key and press F2[MOVE_TO], robot will first move to the base position, and then move to the added positions with joint motion, in displayed order. Please verify that no interference occurs during all motions.

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					Draw No.	B-83744EN/02-03				
02	2019.4.8	Wang	Correction in P6 and P8							
ED	Date	Design	Description							
	Date	2019.1.31	Design	Naitou, Wang	Approve	Arita				

NOTE

For example, when add position 1 and 3 are set to [ENABLED], the robot will move in the following way.

Base position -> translational motion -> base position -> rotational motion -> base position -> add position 1 -> add position 3 -> base position.

- 3 Press [PREV] key to return to the payload identification detail screen.
- 4 Hold down the [SHIFT] key and press F4[EXEC]. The message “Did you verify that no interference occurs during all motions?” will be displayed.
It can also be executed in modes other than AUTO mode. In that case, press the deadman switch and hold down the [SHIFT] key while executing the motion.
- 5 Specify whether to perform payload identification
 - To perform payload identification by running the robot, select [YES].
 - To quit execution, select [NO].

NOTE

- 1 Robot will move. Please adjust the robot motion speed by changing the override.
- 2 If payload setting data while in identify motion is far different from the actual payload, robot may stop by “SYST-320 Program paused by contact stop” or “SYST-325 Payload error is detected”. In that case, please set the payload roughly.

- 6 After the identify motion are finished, payload identification result will be displayed.
- 7 Press F5[APPLY] to set the estimate at a payload setting schedule No. The message “Payload
identification result will be applied. Please apply to DCS parameter after this setup.” will be displayed.
- 8 Specify whether to set the payload identification result.
- To set the payload identification result, select [YES].
 - Not to set the payload identification result, select [NO].
- 9 If the value to be set is greater than the maximum allowable load (indicated in parentheses), the message
“Load is OVER spec! Accept?” will be displayed. Specify whether to set this value, just as in the step
above.

NOTE

- 1 When the payload identification is complete, “*****” of “1 PAYLOAD ESTIMATION *****” changes to [DONE]. However if it is aborted for some reason, such as Emergency Stop, it changes to [ABORT]. The payload are not estimated, so do not [APPLY] and execute the estimation again.
- 2 If the payload identification fails, any or all of the applied results (payload, center of gravity) would be 0.

- 10 Apply DCS parameter. Refer to Dual Check Safety Function OPERATOR'S MANUAL (B-83184EN), Chapter of OVERVIEW, Section of APPLY TO DCS PARAMETER.

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02	2019.4.8	Wang	Correction in P6 and P8			FANUC CORPORATION		Sheet	12／12
ED	Date	Design	Description						
	Date	2019.1.31	Design	Naitou, Wang	Approve				